

Acknowledgement

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Nilay Dhruv

Abstract

The Smart Online Exam Monitoring System is a web-based application developed to conduct online examinations securely. The main purpose of this system is to reduce cheating in online exams by monitoring students in real time using a webcam and browser activity tracking. With the increasing use of online education, maintaining exam integrity has become a major challenge, and this project aims to address that problem.

The system verifies the student's identity using face detection before the exam starts. During the examination, the webcam continuously monitors the student to detect face absence, head movement, looking away from the screen, and face covering. At the same time, the system tracks browser activities such as tab switching or minimizing the exam window. When any suspicious activity is detected, the system generates automatic warnings. If the student violates the rules multiple times, the exam is automatically submitted or terminated.

The admin module allows administrators to upload questions using Excel files, create and schedule exams, and view student activity logs and results. All exam data, logs, and results are stored in a database for future reference. The system is developed using Flask for the backend, HTML, CSS, and JavaScript for the frontend, and MediaPipe for webcam-based face monitoring.

This project provides an automated, lightweight, and cost-effective solution for conducting secure online examinations. It reduces the need for human invigilators and helps educational institutions ensure fairness, honesty, and reliability in online assessment systems.

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